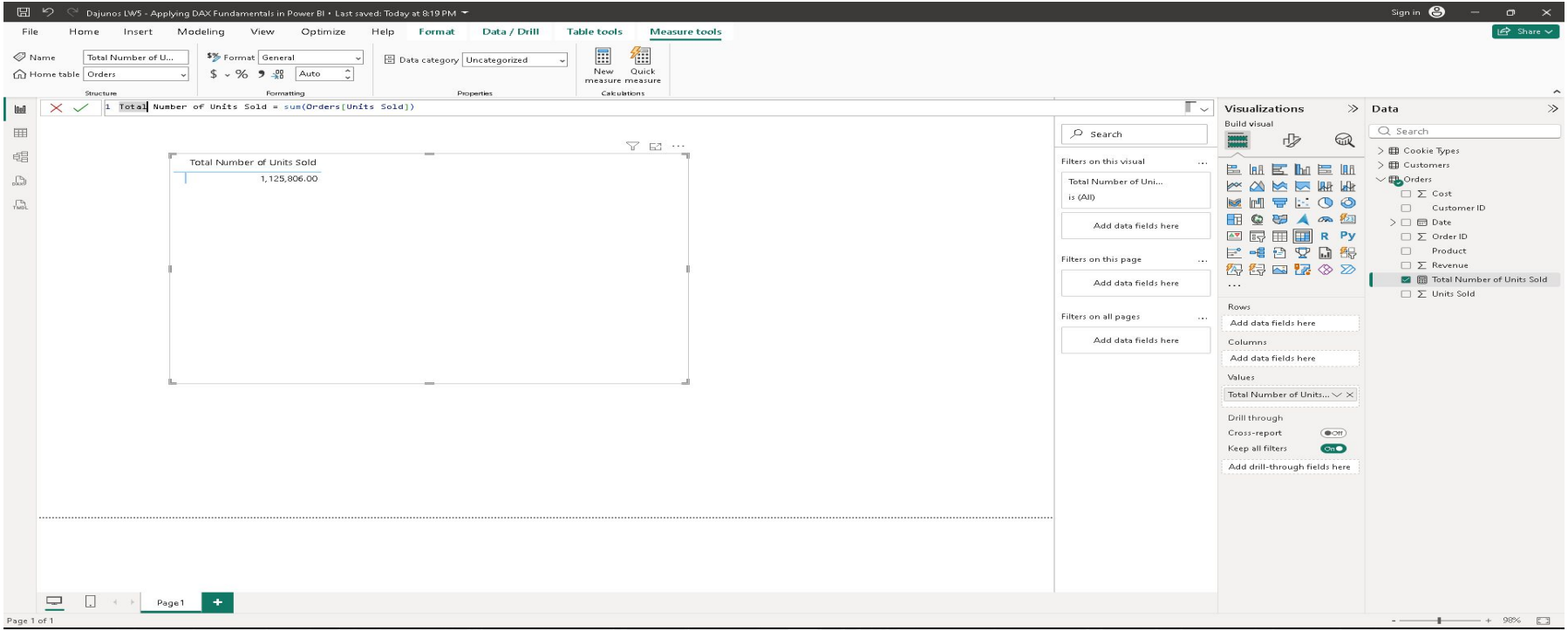
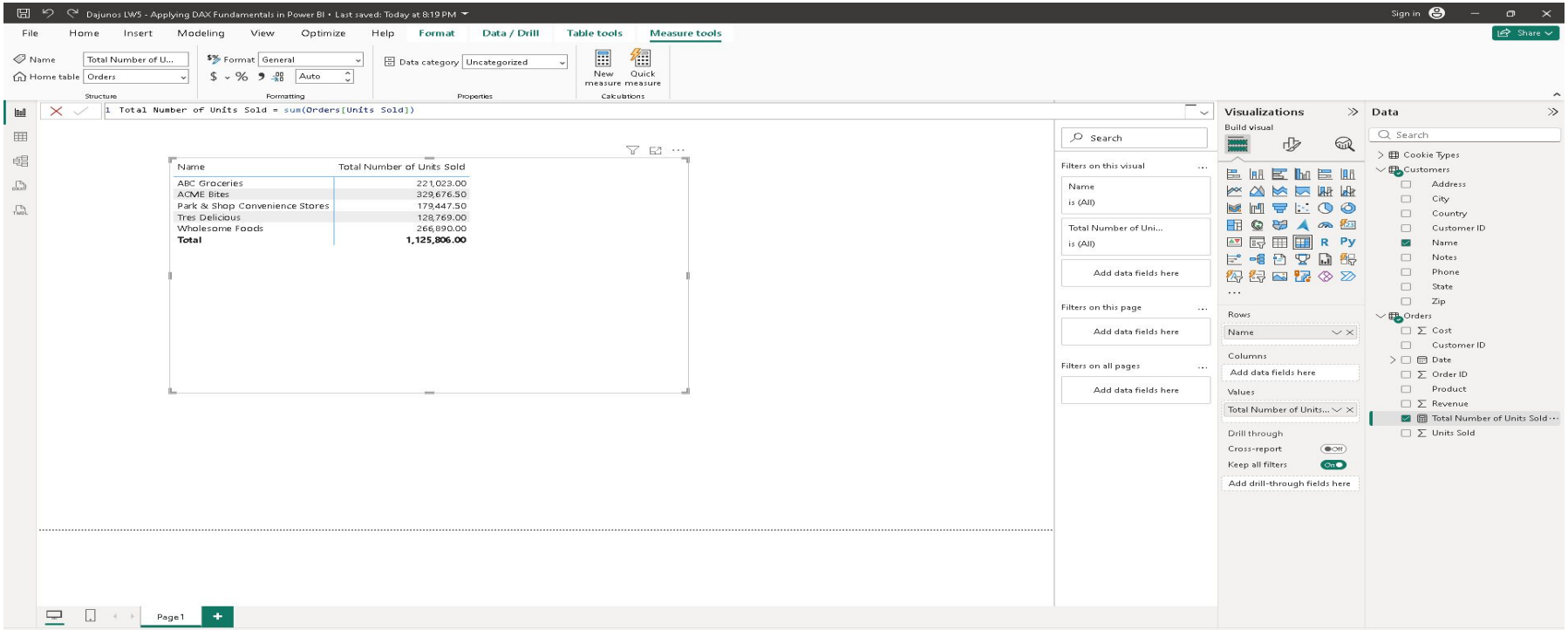
Laboratory Work 5 Activity | Applying DAX Fundamentals in Power BI: From Measures to Contextual Analysis

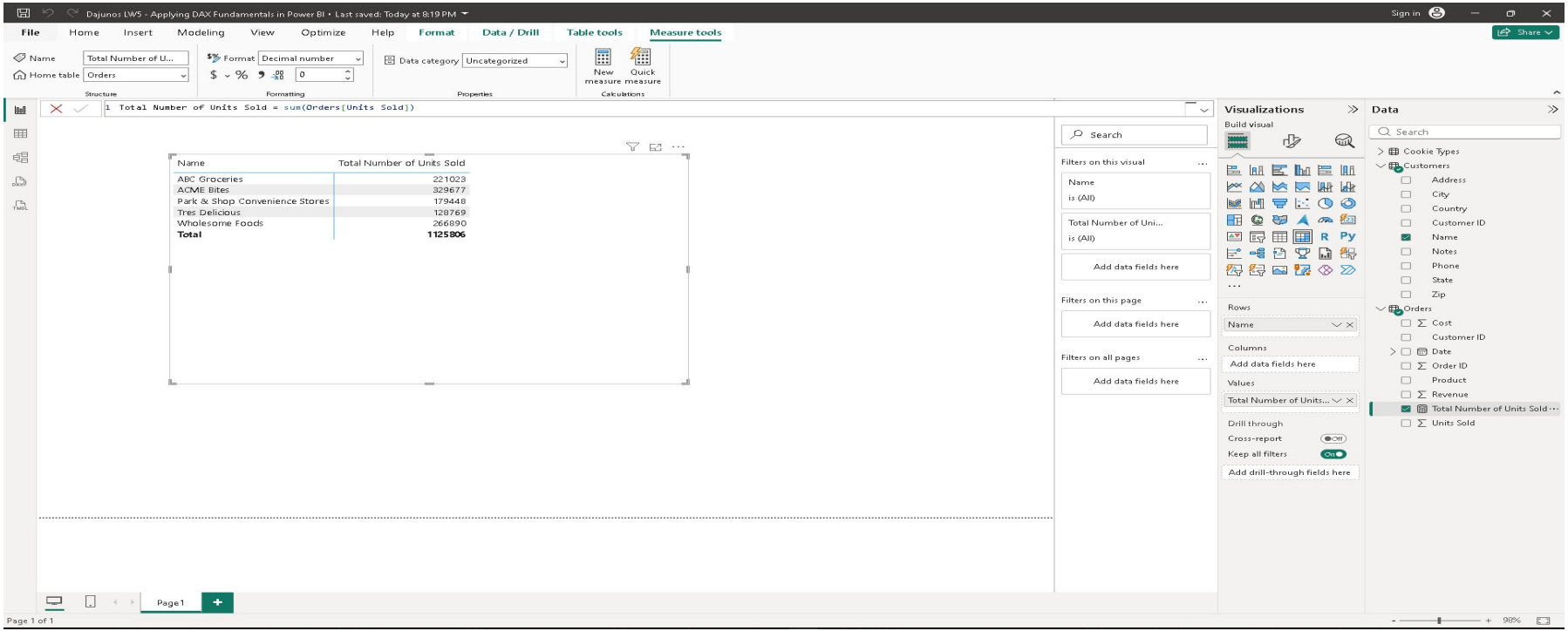
Measure Formulas and Visual Results



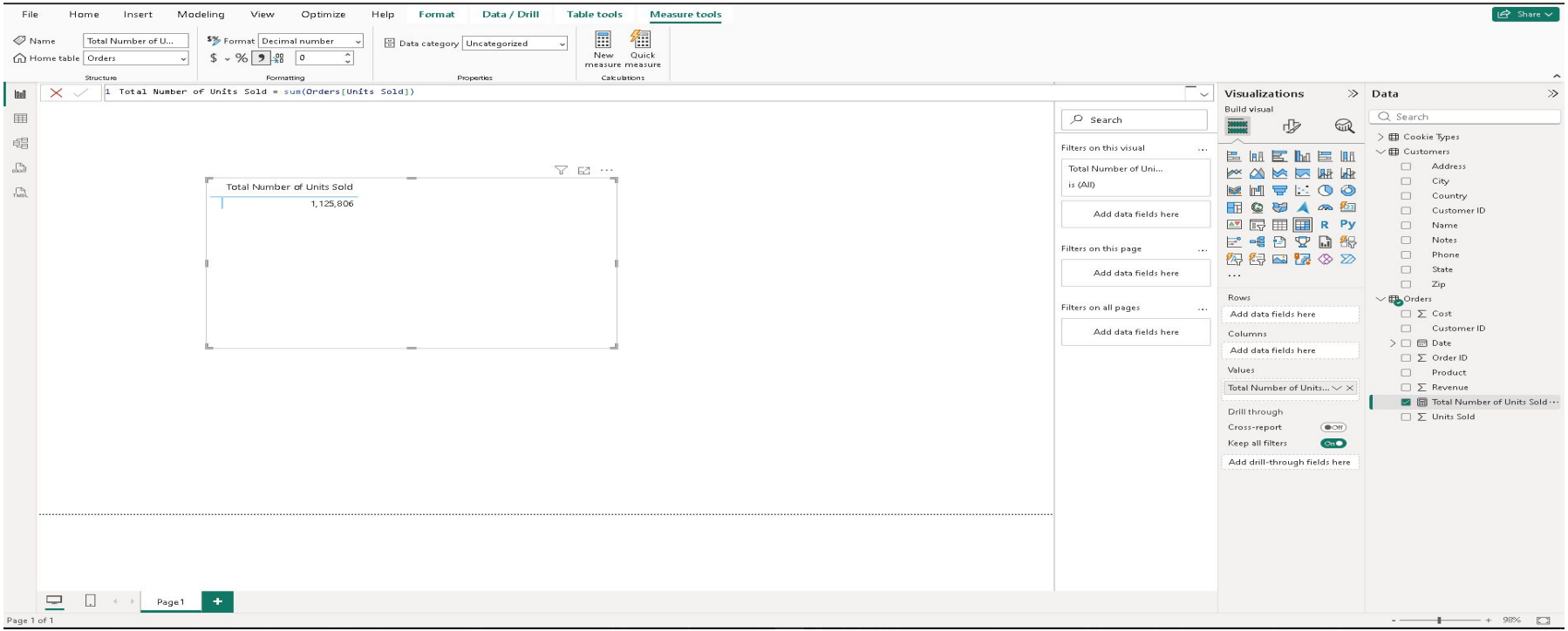
Measure Formulas and Visual Results



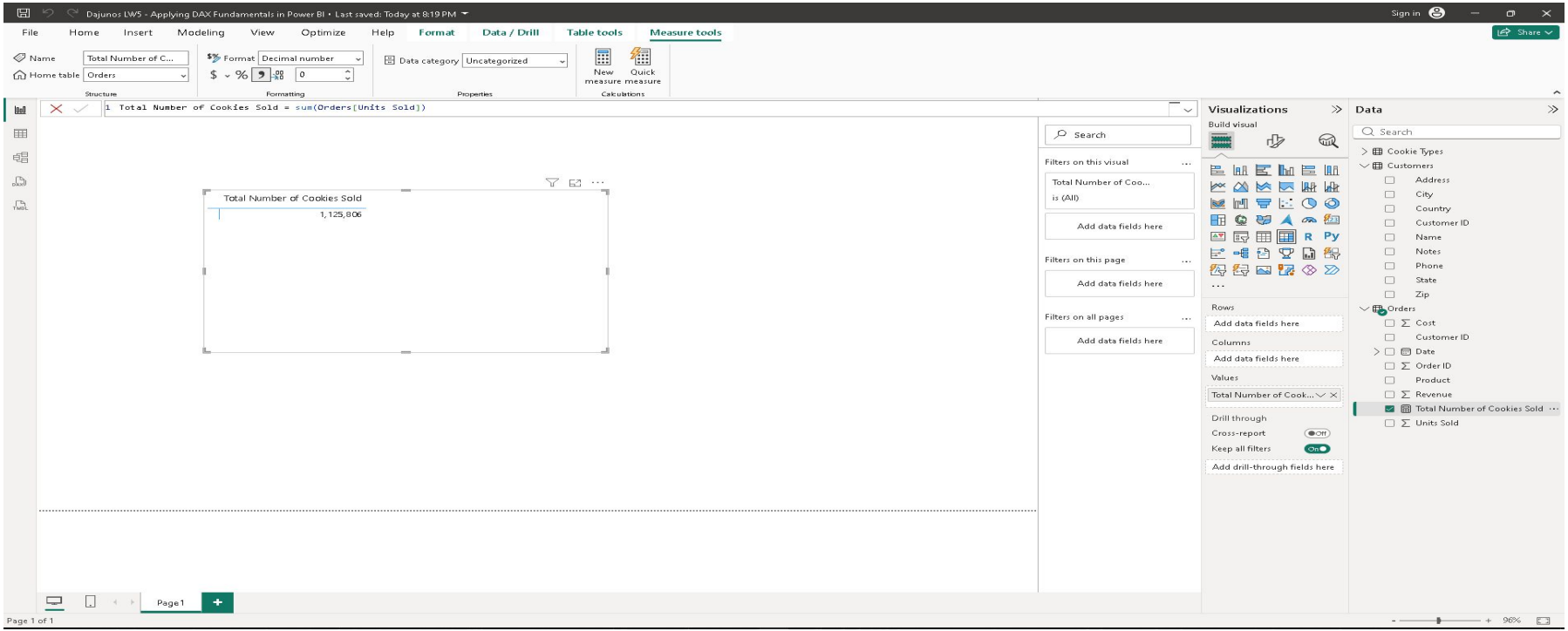
Measure Formulas and Visual Results



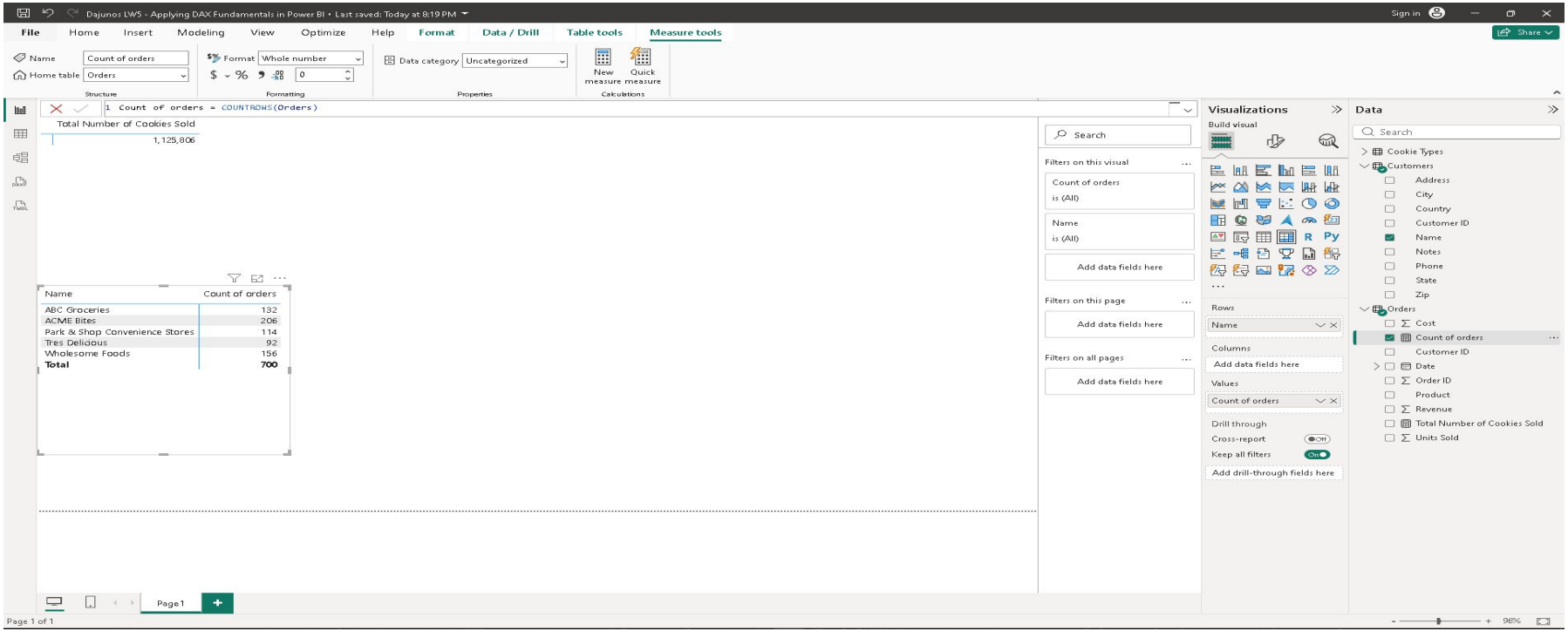
Measure Formulas and Visual Results



Measure Formulas and Visual Results



Measure Formulas and Visual Results



Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools

Name: Distinct Customers
Home table: Orders
Structure

Format: Whole number
Formatting

Data category: Uncategorized
Properties

New measure Quick measure
Calculations

```
1 Distinct Customers = DISTINCTCOUNT(Orders[Customer ID])
```

Total Number of Cookies Sold
1,125,806

Distinct Customers
5

Name	Count of orders
ABC Groceries	132
ACME Bites	206
Park & Shop Convenience Stores	114
Tres Delicious	92
Wholesome Foods	156
Total	**700**

Visualizations
Build visual

Filters on this visual
Distinct Customers is (All)
Add data fields here

Filters on this page
Add data fields here

Filters on all pages
Add data fields here

Rows
Add data fields here
Columns
Add data fields here
Values
Distinct Customers

Drill through
Cross-report Off
Keep all filters On
Add drill-through fields here

Data
Search
- Cookie Types
- Customers
  - Address
  - City
  - Country
  - Customer ID
  - Name
  - Notes
  - Phone
  - State
  - Zip
- Orders
  - Cost
  - Count of orders
  - Customer ID
  - Date
  - Distinct Customers
  - Order ID
  - Product
  - Revenue
  - Total Number of Cookies Sold
  - Units Sold

Page1

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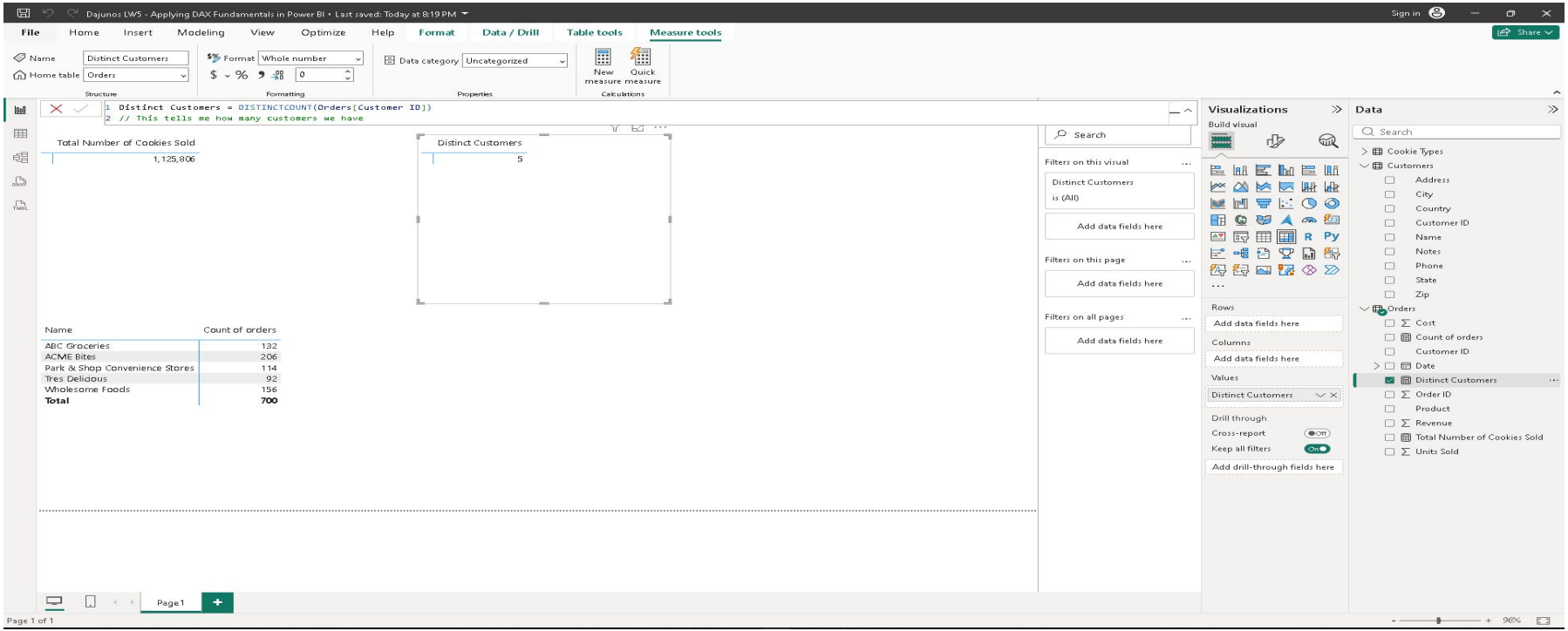
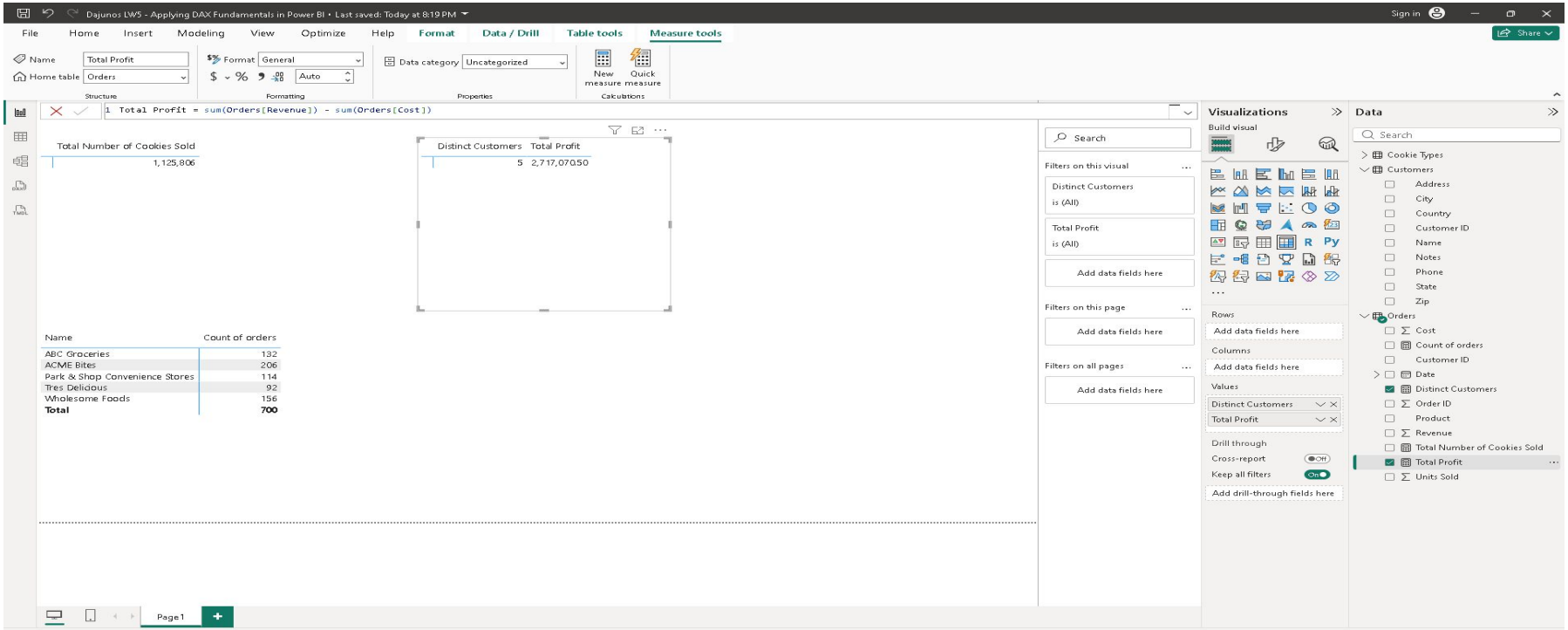
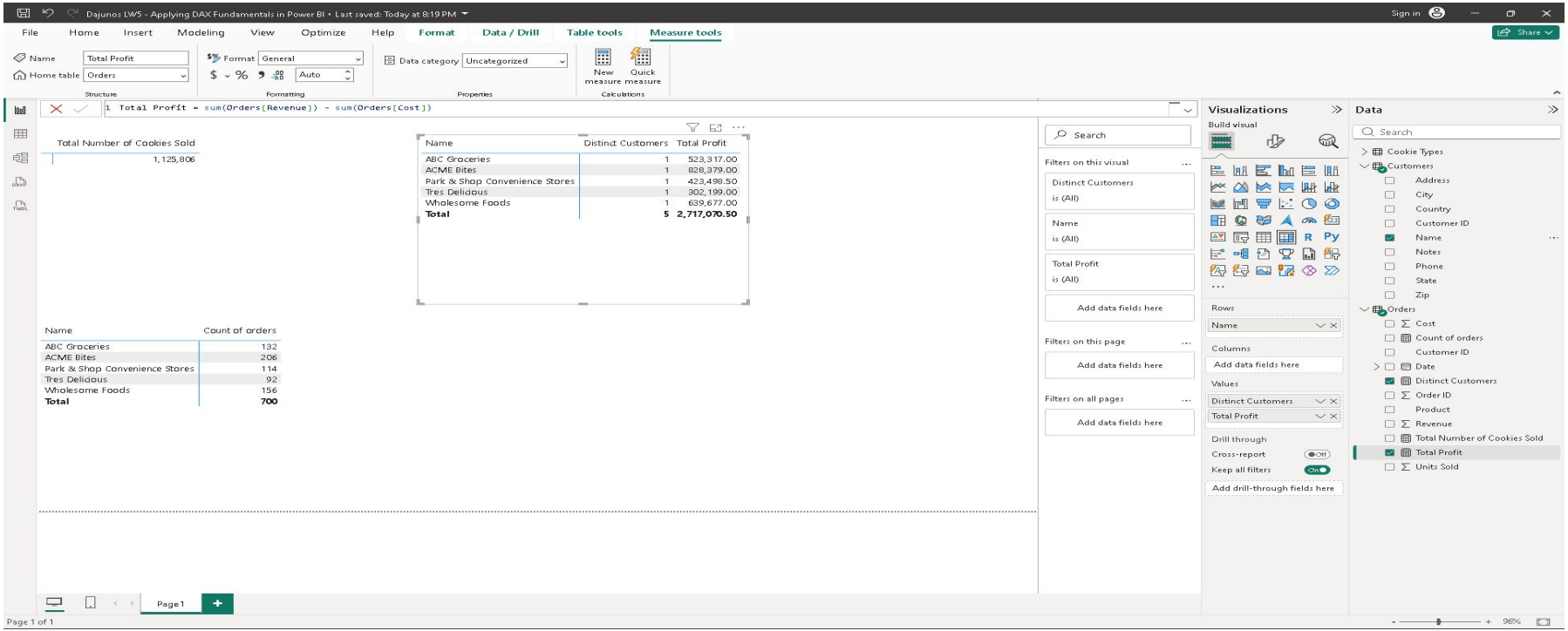
Measure Formulas and Visual Results



Measure Formulas and Visual Results



Measure Formulas and Visual Results



Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

```
Revenue minus Cost =
SUM('Orders'[Revenue]) - SUM('Orders'[Cost])
```

Total Number of Cookies Sold
1,125,806

Name	Distinct Customers	Total Profit	Revenue minus Cost
ABC Groceries	1	523,317.00	523,317.00
ACME Bites	1	828,379.00	828,379.00
Park & Shop Convenience Stores	1	423,498.50	423,498.50
Tres Delicious	1	302,199.00	302,199.00
Wholesome Foods	1	639,677.00	639,677.00
Total	**5**	**2,717,070.50**	**2,717,070.50**

Name	Count of orders
ABC Groceries	132
ACME Bites	206
Park & Shop Convenience Stores	114
Tres Delicious	92
Wholesome Foods	156
Total	**700**

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Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File | Home | Help | Table tools | Column tools

Name: Profit
Data type: Decimal number
Format: General
Auto
Summarization: Sum
Data category: Uncategorized

Structure | Formatting | Properties | Sort | Groups | Relationships | Calculations

Sort by column | Data groups | Manage relationships | New column

```
Profit = Orders[Revenue] - Orders[Cost]
```

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456	3684
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778	4167
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992	4488
3	176392	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598	6897
3	698245	Chocolate Chip	2198	Saturday, August 1, 2020	10990	4396	6594
3	293863	Chocolate Chip	1535	Tuesday, September 1, 2020	7675	3070	4605
3	521535	Chocolate Chip	1404	Friday, November 1, 2019	7020	2808	4212
3	146778	Chocolate Chip	2125	Sunday, December 1, 2019	10625	4250	6375
3	428676	Chocolate Chip	1259	Wednesday, April 1, 2020	6295	2518	3777
3	418690	Chocolate Chip	1366	Monday, June 1, 2020	6830	2732	4098
3	496123	Chocolate Chip	2460	Monday, June 1, 2020	12300	4920	7380
3	513469	Chocolate Chip	1598	Saturday, August 1, 2020	7990	3196	4794
3	231476	Chocolate Chip	2409	Sunday, September 1, 2019	12045	4818	7227
3	100553	Chocolate Chip	1934	Tuesday, September 1, 2020	9670	3868	5802
3	788375	Chocolate Chip	2993	Tuesday, September 1, 2020	14965	5986	8979
3	263663	Chocolate Chip	2146	Friday, November 1, 2019	10730	4292	6438
3	887888	Chocolate Chip	1946	Sunday, December 1, 2019	9730	3892	5838
3	171515	Chocolate Chip	1227	Thursday, October 1, 2020	6135	2454	3681

Table: Orders (700 rows) Column: Profit (657 distinct values)

Data

Search

- Cookie Types
- Customers
- Orders
  - Cost
  - Count of orders
  - Customer ID
  - Date
  - Distinct Customers
  - Order ID
  - Product
  - Profit
  - Revenue
  - Revenue minus Cost
  - Total Number of Cookies Sold
  - Total Profit
  - Units Sold

Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File | Home | Insert | Modeling | View | Optimize | Help | Format | Data / Drill | Table tools | Measure tools

Name: Revenue minus Cost
Home table: Orders
Format: General
Auto
Data category: Uncategorized
New measure | Quick measure

Structure | Formatting | Properties | Calculations

```
1 Revenue minus Cost =
2 SUM('Orders'[Revenue]) - SUM('Orders'[Cost])
```

Total Number of Cookies Sold
1,125,806

Cookie Type	Distinct Customers	Total Profit	Revenue minus Cost
Chocolate Chip	5	1,014,718.50	1,014,718.50
Fortune Cookie	5	77,100.00	77,100.00
Oatmeal Raisin	5	434,881.00	434,881.00
Snickerdoodle	5	367,118.00	367,118.00
Sugar	5	295,373.00	295,373.00
White Chocolate Macadamia Nut	5	527,880.00	527,880.00
Total	**5**	**2,717,070.50**	**2,717,070.50**

Name	Count of orders
ABC Groceries	132
ACME Bites	206
Park & Shop Convenience Stores	114
Tres Delicious	92
Wholesome Foods	156
Total	**700**

Cookie Type	Profit Margin %
Chocolate Chip	60%
Fortune Cookie	50%
Oatmeal Raisin	56%
Snickerdoodle	63%
Sugar	58%
White Chocolate Macadamia Nut	54%
Total	**58%**

Search

Filters on this visual
Cookie Type is (All)
Distinct Customers is (All)
Revenue minus Cost is (All)
Total Profit is (All)
Add data fields here

Filters on this page
Add data fields here

Filters on all pages
Add data fields here

Quick measure

Visualizations
Build visual

Rows: Cookie Type
Columns: Add data fields here
Values: Distinct Customers, Total Profit, Revenue minus Cost

Drill through
Cross-report: Off
Keep all filters: On
Add drill-through fields here

Data
Search

- Cookie Types
  - Cookie Type
  - Cost Per Cookie
  - Revenue Per Cookie
  - Units Sold
- Customers
  - Address
  - City
  - Country
  - Customer ID
  - Name
  - Notes
  - Phone
  - State
  - Zip
- Orders
  - Cost
  - Count of orders
  - Customer ID
  - Date
  - Distinct Customers
  - Order ID
  - Product
  - Profit
  - Profit Margin %
  - Revenue
  - Revenue minus Cost
  - Total Number of Cookies Sold
  - Total Profit
  - Units Sold

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Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools

Name: Profit Margin %
Home table: Orders
Structure

Format: Percentage
Formatting

Data category: Uncategorized
Properties

New measure | Quick measure
Calculations

```
1 Profit Margin % = [Total Profit] / sum(Orders[Revenue])
```

Total Number of Cookies Sold
1,125,806

Cookie Type	Distinct Customers	Total Profit	Revenue minus Cost
Chocolate Chip	5	1,014,718.50	1,014,718.50
Fortune Cookie	5	77,100.00	77,100.00
Oatmeal Raisin	5	434,881.00	434,881.00
Snickerdoodle	5	367,118.00	367,118.00
Sugar	5	295,373.00	295,373.00
White Chocolate Macadamia Nut	5	527,880.00	527,880.00
Total	**5**	**2,717,070.50**	**2,717,070.50**

Name	Count of orders
ABC Groceries	132
ACME Bites	206
Park & Shop Convenience Stores	114
Tres Delicious	92
Wholesome Foods	156
Total	**700**

Cookie Type	Profit Margin %
Chocolate Chip	60%
Fortune Cookie	50%
Oatmeal Raisin	56%
Snickerdoodle	63%
Sugar	58%
White Chocolate Macadamia Nut	54%
Total	**58%**

Filters on this visual
Cookie Type is (All)
Profit Margin % is (All)
Add data fields here

Filters on this page
Add data fields here

Filters on all pages
Add data fields here

Visualizations
Build visual

Rows: Cookie Type
Columns: Add data fields here
Values: Profit Margin %
Drill through
Cross-report: Off
Keep all filters: On
Add drill-through fields here

Data
Cookie Types
- Cookie Type
- Cost Per Cookie
- Revenue Per Cookie
- Units Sold

Customers
- Address
- City
- Country
- Customer ID
- Name
- Notes
- Phone
- State
- Zip

Orders
- Cost
- Count of orders
- Customer ID
- Date
- Distinct Customers
- Order ID
- Product
- Profit
- Profit Margin %
- Revenue
- Revenue minus Cost
- Total Number of Cookies Sold
- Total Profit
- Units Sold

Page1

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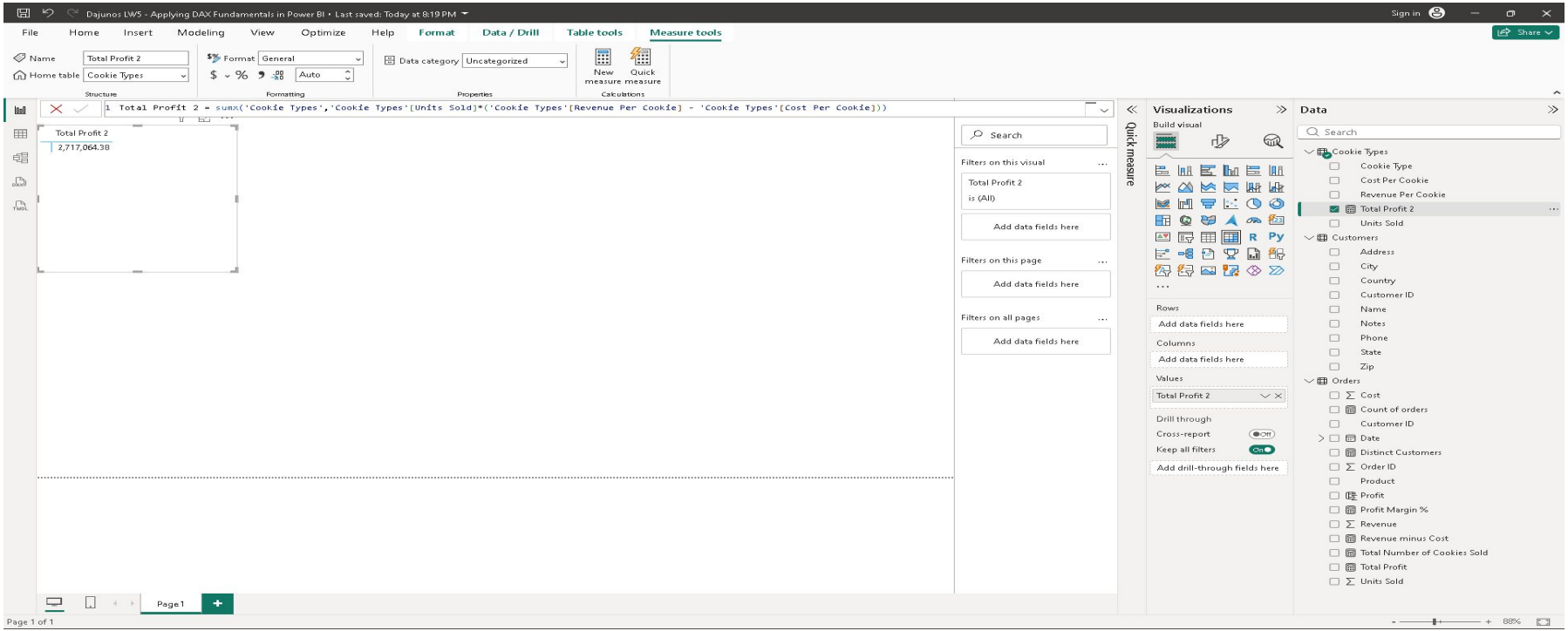
Measure Formulas and Visual Results



Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File | Home | Help | Table tools | Column tools

Name: Day of week
Data type: Whole number
Format: Whole number
Summarization: Sum
Data category: Uncategorized

```
1 Day of week = WEEKDAY(Orders[Date],1)
```

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of week
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456	3684	3
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778	4167	3
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992	4488	2
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598	6897	3
3	698245	Chocolate Chip	2198	Saturday, August 1, 2020	10990	4396	6594	7
3	293863	Chocolate Chip	1535	Tuesday, September 1, 2020	7675	3070	4605	3
3	521535	Chocolate Chip	1404	Friday, November 1, 2019	7020	2808	4212	6
3	146778	Chocolate Chip	2125	Sunday, December 1, 2019	10625	4250	6375	1
3	428676	Chocolate Chip	1259	Wednesday, April 1, 2020	6295	2518	3777	4
3	418690	Chocolate Chip	1366	Monday, June 1, 2020	6830	2732	4098	2
3	496123	Chocolate Chip	2460	Monday, June 1, 2020	12300	4920	7380	2
3	513469	Chocolate Chip	1598	Saturday, August 1, 2020	7990	3196	4794	7
3	231476	Chocolate Chip	2409	Sunday, September 1, 2019	12045	4818	7227	1
3	100553	Chocolate Chip	1934	Tuesday, September 1, 2020	9670	3868	5802	3
3	788375	Chocolate Chip	2993	Tuesday, September 1, 2020	14965	5986	8979	3
3	263663	Chocolate Chip	2146	Friday, November 1, 2019	10730	4292	6438	6
3	887888	Chocolate Chip	1946	Sunday, December 1, 2019	9730	3892	5838	1
3	171515	Chocolate Chip	1227	Thursday, October 1, 2020	6135	2454	3681	5

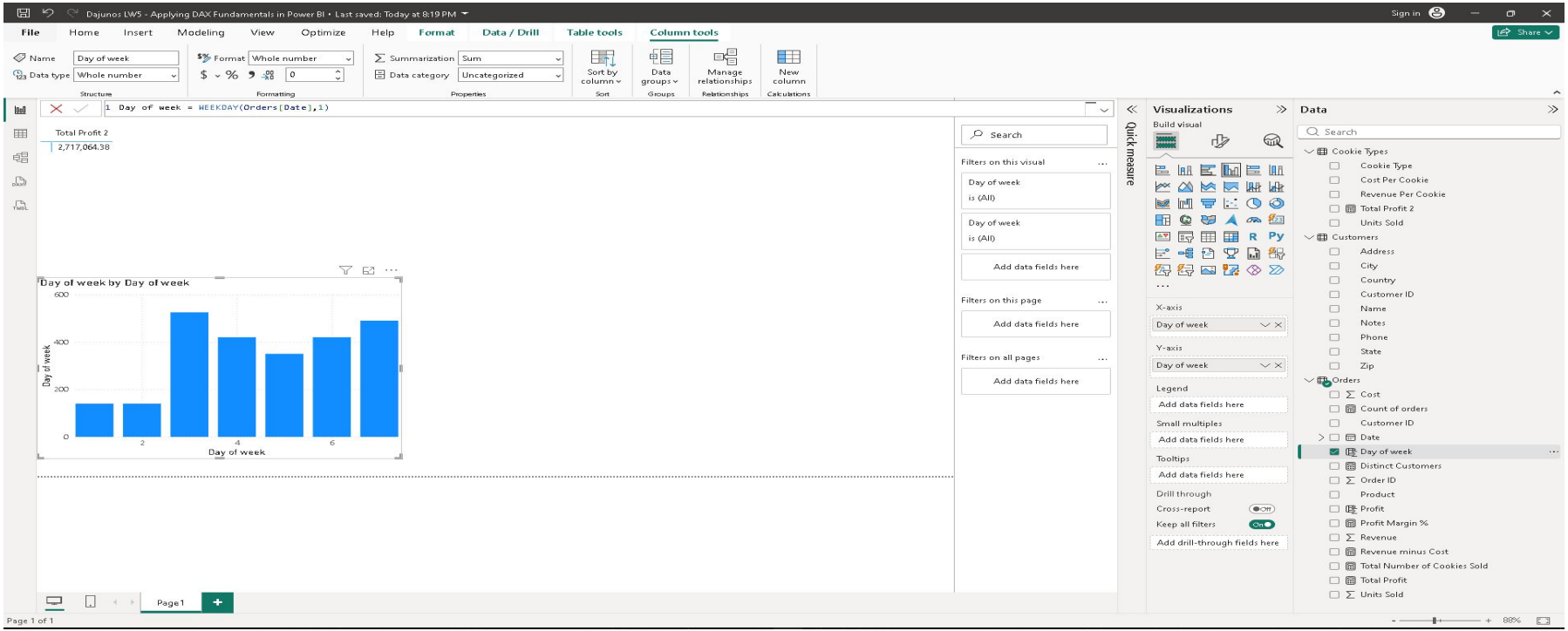
Table: Orders (700 rows) Column: Day of week (7 distinct values)

Data

Search

- Cookie Types
  - Cookie Type
  - Cost Per Cookie
  - Revenue Per Cookie
  - Total Profit 2
  - Units Sold
- Customers
- Orders
  - Cost
  - Count of orders
  - Customer ID
  - Date
  - Day of week
  - Distinct Customers
  - Order ID
  - Product
  - Profit
  - Profit Margin %
  - Revenue
  - Revenue minus Cost
  - Total Number of Cookies Sold
  - Total Profit
  - Units Sold

Measure Formulas and Visual Results



Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File | Home | Help | Table tools | Column tools

Name: Orders

Structure | Relationships: Manage relationships | Calculations: New measure, Quick measure, New column, New table | Calendars: Mark as date table

```
1 Has Chocolate = if(FIND("Chocolate",Orders[Product],1,0)>0,"Has Chocolate","No Chocolate")
```

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of week	Has Chocolate
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7	Has Chocolate
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7	Has Chocolate
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2	Has Chocolate
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3	Has Chocolate
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6	Has Chocolate
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7	Has Chocolate
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2	Has Chocolate
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6	Has Chocolate
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3	Has Chocolate
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3	Has Chocolate
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2	Has Chocolate
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2	Has Chocolate
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4	Has Chocolate
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6	Has Chocolate
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4	Has Chocolate
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3	Has Chocolate
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3	Has Chocolate
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3	Has Chocolate
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3	Has Chocolate
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1	Has Chocolate
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456	3684	3	Has Chocolate
3	691337	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778	4167	3	Has Chocolate
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992	4488	2	Has Chocolate
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598	6897	3	Has Chocolate
3	698245	Chocolate Chip	2198	Saturday, August 1, 2020	10990	4396	6594	7	Has Chocolate
3	293863	Chocolate Chip	1535	Tuesday, September 1, 2020	7675	3070	4605	3	Has Chocolate
3	521535	Chocolate Chip	1404	Friday, November 1, 2019	7020	2808	4212	6	Has Chocolate
3	146778	Chocolate Chip	2125	Sunday, December 1, 2019	10625	4250	6375	1	Has Chocolate
3	428676	Chocolate Chip	1259	Wednesday, April 1, 2020	6295	2518	3777	4	Has Chocolate
3	418690	Chocolate Chip	1366	Monday, June 1, 2020	6830	2732	4098	2	Has Chocolate
3	496123	Chocolate Chip	2460	Monday, June 1, 2020	12300	4920	7380	2	Has Chocolate
3	513469	Chocolate Chip	1598	Saturday, August 1, 2020	7990	3196	4794	7	Has Chocolate
3	231476	Chocolate Chip	2409	Sunday, September 1, 2019	12045	4818	7227	1	Has Chocolate
3	100553	Chocolate Chip	1934	Tuesday, September 1, 2020	9670	3868	5802	3	Has Chocolate
3	788375	Chocolate Chip	2993	Tuesday, September 1, 2020	14965	5986	8979	3	Has Chocolate
3	263663	Chocolate Chip	2146	Friday, November 1, 2019	10730	4292	6438	6	Has Chocolate
3	887888	Chocolate Chip	1946	Sunday, December 1, 2019	9730	3892	5838	1	Has Chocolate
3	171515	Chocolate Chip	1227	Thursday, October 1, 2020	6135	2454	3681	5	Has Chocolate

Table: Orders (700 rows) Column: Has Chocolate (2 distinct values)

Data

Search

- Cookie Types
  - Cookie Type
  - Cost Per Cookie
  - Revenue Per Cookie
  - Total Profit 2
  - Units Sold
- Customers
- Orders
  - Cost
  - Count of orders
  - Customer ID
  - Date
  - Day of week
  - Distinct Customers
  - Has Chocolate
  - Order ID
  - Product
  - Profit
  - Profit Margin %
  - Revenue
  - Revenue minus Cost
  - Total Number of Cookies Sold
  - Total Profit
  - Units Sold

Measure Formulas and Visual Results

Dajunos LW5 - Applying DAX Fundamentals in Power BI • Last saved: Today at 8:19 PM

File | Home | Help | Table tools | Measure tools

Name: Chocolate chip wit...
Home table: Orders
Structure

Format: Whole number
Formatting

Data category: Uncategorized
Properties

New measure | Quick measure
Calculations

```
1 Chocolate chip with over 500 units = CALCULATE(COUNTROWS(Orders),Orders[Units Sold]>500,Orders[Product]=="Chocolate Chip")
```

Customer ID	Order ID	Product	Units Sold	Date	Revenue	Cost	Profit	Day of week	Has Chocolate
3	266868	Chocolate Chip	292	Saturday, February 1, 2020	1460	584	876	7	Has Chocolate
3	140794	Chocolate Chip	974	Saturday, February 1, 2020	4870	1948	2922	7	Has Chocolate
3	684759	Chocolate Chip	2518	Monday, June 1, 2020	12590	5036	7554	2	Has Chocolate
3	183251	Chocolate Chip	1513	Tuesday, December 1, 2020	7565	3026	4539	3	Has Chocolate
3	600124	Chocolate Chip	1030	Friday, May 1, 2020	5150	2060	3090	6	Has Chocolate
3	562219	Chocolate Chip	1514	Saturday, February 1, 2020	7570	3028	4542	7	Has Chocolate
3	387444	Chocolate Chip	787	Monday, June 1, 2020	3935	1574	2361	2	Has Chocolate
3	365463	Chocolate Chip	1728	Friday, May 1, 2020	8640	3456	5184	6	Has Chocolate
3	505159	Chocolate Chip	2145	Tuesday, October 1, 2019	10725	4290	6435	3	Has Chocolate
3	300303	Chocolate Chip	1084	Tuesday, December 1, 2020	5420	2168	3252	3	Has Chocolate
3	578401	Chocolate Chip	689	Monday, June 1, 2020	3445	1378	2067	2	Has Chocolate
3	365552	Chocolate Chip	1570	Monday, June 1, 2020	7850	3140	4710	2	Has Chocolate
3	239419	Chocolate Chip	4251	Wednesday, January 1, 2020	21255	8502	12753	4	Has Chocolate
3	549329	Chocolate Chip	2918	Friday, May 1, 2020	14590	5836	8754	6	Has Chocolate
3	288851	Chocolate Chip	2988	Wednesday, July 1, 2020	14940	5976	8964	4	Has Chocolate
3	255145	Chocolate Chip	2074	Tuesday, September 1, 2020	10370	4148	6222	3	Has Chocolate
3	538134	Chocolate Chip	1514	Tuesday, October 1, 2019	7570	3028	4542	3	Has Chocolate
3	817134	Chocolate Chip	274	Tuesday, December 1, 2020	1370	548	822	3	Has Chocolate
3	697568	Chocolate Chip	1138	Tuesday, December 1, 2020	5690	2276	3414	3	Has Chocolate
3	508782	Chocolate Chip	2992	Sunday, March 1, 2020	14960	5984	8976	1	Has Chocolate
3	697895	Chocolate Chip	1228	Tuesday, October 1, 2019	6140	2456	3684	3	Has Chocolate
3	691331	Chocolate Chip	1389	Tuesday, October 1, 2019	6945	2778	4167	3	Has Chocolate
3	568366	Chocolate Chip	1496	Monday, June 1, 2020	7480	2992	4488	2	Has Chocolate
3	176592	Chocolate Chip	2299	Tuesday, October 1, 2019	11495	4598	6897	3	Has Chocolate
3	698245	Chocolate Chip	2198	Saturday, August 1, 2020	10990	4396	6594	7	Has Chocolate
3	293863	Chocolate Chip	1535	Tuesday, September 1, 2020	7675	3070	4605	3	Has Chocolate
3	521535	Chocolate Chip	1404	Friday, November 1, 2019	7020	2808	4212	6	Has Chocolate
3	146778	Chocolate Chip	2125	Sunday, December 1, 2019	10625	4250	6375	1	Has Chocolate
3	428676	Chocolate Chip	1259	Wednesday, April 1, 2020	6295	2518	3777	4	Has Chocolate
3	418690	Chocolate Chip	1366	Monday, June 1, 2020	6830	2732	4098	2	Has Chocolate
3	496123	Chocolate Chip	2460	Monday, June 1, 2020	12300	4920	7380	2	Has Chocolate
3	513469	Chocolate Chip	1598	Saturday, August 1, 2020	7990	3196	4794	7	Has Chocolate
3	231476	Chocolate Chip	2409	Sunday, September 1, 2019	12045	4818	7227	1	Has Chocolate
3	100553	Chocolate Chip	1934	Tuesday, September 1, 2020	9670	3868	5802	3	Has Chocolate
3	788375	Chocolate Chip	2993	Tuesday, September 1, 2020	14965	5986	8979	3	Has Chocolate
3	263663	Chocolate Chip	2146	Friday, November 1, 2019	10730	4292	6438	6	Has Chocolate
3	887888	Chocolate Chip	1946	Sunday, December 1, 2019	9730	3892	5838	1	Has Chocolate
3	171515	Chocolate Chip	1227	Thursday, October 1, 2020	6135	2454	3681	5	Has Chocolate

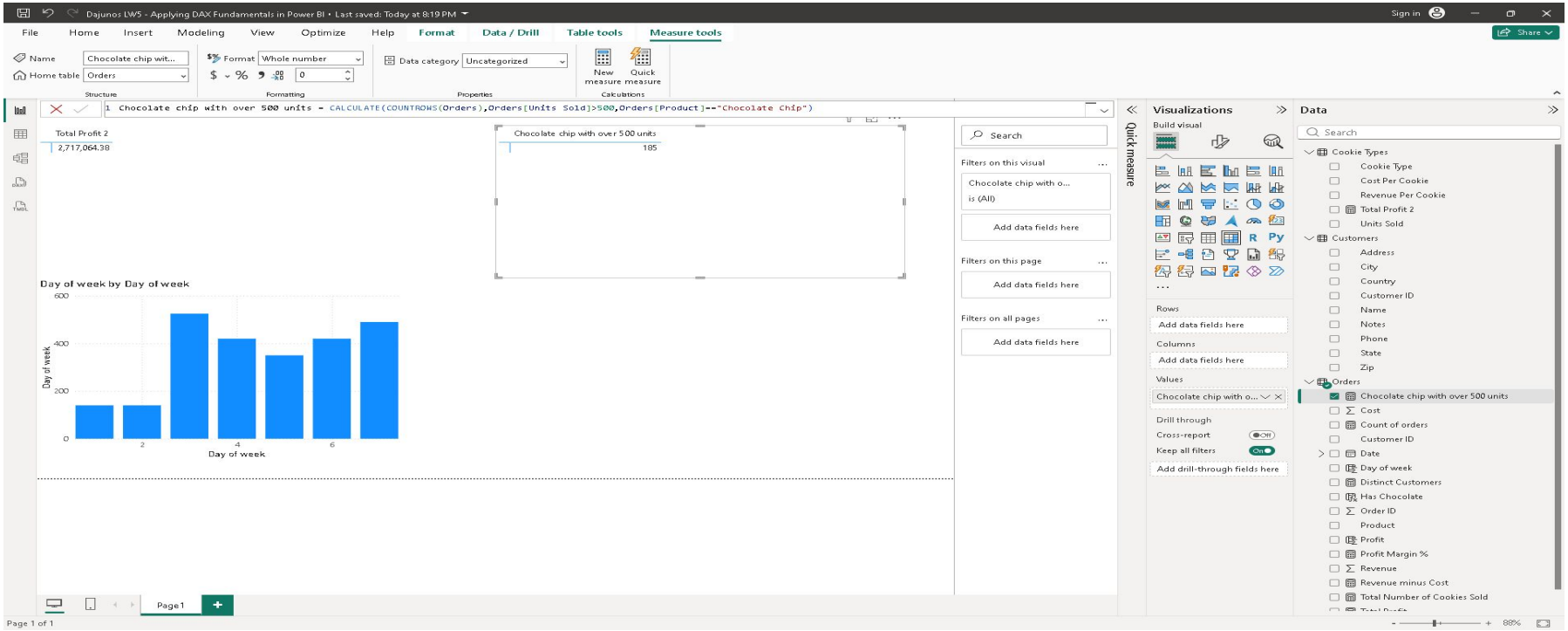
Table: Orders (700 rows) Column: Chocolate chip with over 500 units (0 distinct values)

Data

Search

- Cookie Types
  - Cookie Type
  - Cost Per Cookie
  - Revenue Per Cookie
  - Total Profit 2
  - Units Sold
- Customers
- Orders
  - Chocolate chip with over...
  - Cost
  - Count of orders
  - Customer ID
  - Date
  - Day of week
  - Distinct Customers
  - Has Chocolate
  - Order ID
  - Product
  - Profit
  - Profit Margin %
  - Revenue
  - Revenue minus Cost
  - Total Number of Cookies Sold
  - Total Profit
  - Units Sold

Measure Formulas and Visual Results



Guide Questions:

I. Foundational Concepts

1. Defining DAX:

An Excel formula works like a calculator for cells, you point it to a specific cell like =A1+B1 and it gives you a result. It only looks at the cells you tell it to. DAX is different because it works with whole tables and relationships between tables. Instead of pointing to a cell, you point to a column like SUM(Sales[Sales Amount]). The big difference is that DAX is smart about filters, the same formula can give different answers depending on what you click or filter in your report. Excel formulas are fixed; DAX formulas are flexible and dynamic.

Guide Questions:

I. Foundational Concepts

2. The “Why” of DAX:

Raw data is just a list of numbers it can't do anything on its own. If you want to know the total sales for just one product category in one specific year, the raw data can't figure that out automatically. DAX fills that gap by letting you create calculations that update automatically based on what the user is looking at. When you click on a filter or a chart, DAX recalculates instantly. It turns plain data into useful, interactive answers.

Guide Questions:

II. Calculated Columns vs. Measures

3. Storage Difference:

Calculated Columns make the file bigger. This is because when you create a Calculated Column, Power BI computes a value for every single row in your table and then saves all those results inside the file. The more rows you have, the bigger the file gets. Measures do not increase the file size because they are only calculated when you need them like when a chart loads. Once the chart is done, the result is thrown away and not saved. So for big datasets, it is smarter to use Measures whenever possible.

Guide Questions:

II. Calculated Columns vs. Measures

4. Context Clues:

Filter context simply means what is currently selected or filtered in your report like a slicer set to "2024" or a chart showing only "Chocolate Chip" cookies. A Measure has no fixed answer. It recalculates based on whatever filter is active. For example, a measure for Total Sales might show ₱500,000 for all products, but if you click on "Chocolate Chip" in a chart, that same measure will now show only the sales for Chocolate Chip maybe ₱120,000 without changing the formula at all. The filter context controls what data the Measure sees.

Guide Questions:

III. Function Application & Syntax

5. The RELATED Function:

When your data is spread across multiple tables, you sometimes need to grab a value from another table. The RELATED() function lets you do that it pulls a value from a connected table into your current table. But for it to work, you first need to create a relationship between the two tables in the Model View. Think of it like connecting two Lego pieces without the connection, RELATED has no path to follow and will give you an error. For example, the Orders table and the Customers table must be linked by a shared column like CustomerID.

Guide Questions:

III. Function Application & Syntax

6. Deconstructing the Formula: Total Sales = SUM(Sales[Sales Amount])

Total Sales - name of the measure

SUM() - The function - it adds up all the values

Sales - The table where the data lives

Sales Amount - The column being added up

Guide Questions:

IV. Analysis & Troubleshooting

7. Data Interpretation:

No, it is not broken. A negative profit just means the costs were higher than the revenue for that category. The formula is working correctly, it is simply showing you a bad business result, not a bad formula.

Guide Questions:

IV. Analysis & Troubleshooting

8. Optimization:

A Measure is the much better choice. Here is why in simple terms: A Calculated Column would compute and save a profit margin value for every single row in your table. If you have 100,000 rows, that is 100,000 stored values taking up space even if you only ever look at the total. A Measure only calculates when a visual needs it, and it responds to whatever filter is active. It is fast, lightweight, and gives you the right answer based on what is selected. Since Profit Margin Percentage is something you look at as a summary or KPI (not something you need per individual row), a Measure is the smarter and more efficient option. It keeps your file small and your report fast.